

Daksh — Class X Study Test

Subject: Mathematics | **Test:** Test 1 (Day 7, 16 May 2026) — Weekly

Time Allowed: 3 Hours • **Maximum Marks:** 80

Syllabus covered: Real Numbers • Polynomials • Pair of Linear Equations in Two Variables • Quadratic Equations • Arithmetic Progressions • Triangles

General Instructions:

- (i) This question paper contains five sections — A, B, C, D and E.
- (ii) Section A has Multiple Choice / Very Short Answer Questions of 1 mark each.
- (iii) Section B has Short Answer Questions of 2 marks each.
- (iv) Section C has Short Answer Questions of 3 marks each.
- (v) Section D has Long Answer Questions of 5 marks each.
- (vi) Section E has Case-based / Source-based questions of 4 marks each.
- (vii) All questions are compulsory. Internal choice has been provided in some questions wherever indicated.
- (viii) Use of calculators is not permitted.

SECTION A — Multiple Choice / Very Short Answer Questions (20 × 1 = 20 marks)

Choose the correct option / answer in one word or one sentence.

- Q1.** The HCF of 96 and 404 by the prime-factorisation method is: (a) 2 (b) 4 (c) 8 (d) 12
- Q2.** If two positive integers a and b are written as $a = x^3y^2$ and $b = xy^3$, where x, y are prime numbers, then $\text{HCF}(a,b)$ is: (a) xy (b) xy^2 (c) x^3y^3 (d) x^2y^2
- Q3.** The decimal expansion of $23/(2^3 \times 5^2)$ will terminate after how many decimal places? (a) 1 (b) 2 (c) 3 (d) 4
- Q4.** The number of zeroes a quadratic polynomial can have at most is: (a) 0 (b) 1 (c) 2 (d) 3
- Q5.** If α and β are zeroes of the polynomial $x^2 - 5x + 6$, then $\alpha + \beta = ?$ (a) 5 (b) -5 (c) 6 (d) -6
- Q6.** The pair of equations $x + 2y = 5$ and $2x + 4y = 10$ has: (a) unique solution (b) no solution (c) infinitely many solutions (d) two solutions
- Q7.** If the pair of linear equations is consistent and has a unique solution, the lines representing them are: (a) parallel (b) intersecting (c) coincident (d) perpendicular only
- Q8.** The roots of the equation $x^2 - 3x - 10 = 0$ are: (a) 5, -2 (b) -5, 2 (c) 5, 2 (d) -5, -2
- Q9.** The discriminant of $2x^2 - 4x + 3 = 0$ is: (a) 8 (b) -8 (c) 4 (d) -4
- Q10.** A quadratic equation with real coefficients has equal roots when its discriminant is: (a) positive (b) zero (c) negative (d) any value
- Q11.** The 10th term of the AP: 2, 7, 12, 17, ... is: (a) 47 (b) 50 (c) 52 (d) 45
- Q12.** The common difference of the AP $1/p, (1-p)/p, (1-2p)/p, \dots$ is: (a) 1 (b) $1/p$ (c) -1 (d) $-1/p$
- Q13.** The sum of the first 20 natural numbers is: (a) 200 (b) 210 (c) 220 (d) 190
- Q14.** If in two triangles ABC and PQR, $AB/PQ = BC/PR = CA/RQ$, then: (a) $\triangle PQR \sim \triangle CAB$ (b) $\triangle PQR \sim \triangle ABC$ (c) $\triangle CBA \sim \triangle PQR$ (d) $\triangle BCA \sim \triangle PQR$
- Q15.** In $\triangle ABC$, DE $\parallel$ BC. If AD = 1.5 cm, DB = 3 cm and AE = 1 cm, then EC = ? (a) 1 cm (b) 2 cm (c) 3 cm (d) 1.5 cm
- Q16.** Two triangles are similar if their corresponding sides are: (a) equal (b) proportional (c) perpendicular (d) parallel
- Q17.** If $65 = 5 \times q + r$, where $0 \leq r < 5$, then $q = ?$ (Use Euclid's algorithm.) (Answer in one word.)
- Q18.** Write whether $\sqrt{2}$ is rational or irrational.
- Q19.** State the Basic Proportionality Theorem (BPT) in one sentence.
- Q20.** If α, β are zeroes of $x^2 + 5x + 6$, find $\alpha\beta$.

SECTION B — Short Answer-I (5 × 2 = 10 marks)

Answer each question in 30–50 words / show working.

- Q21.** Find the LCM and HCF of 26 and 91 using the fundamental theorem of arithmetic, and verify that $\text{LCM} \times \text{HCF} = \text{product of the two numbers}$.
- Q22.** Find a quadratic polynomial whose zeroes are 3 and -2 . Verify the relationship between zeroes and coefficients.
- Q23.** Solve for x and y : $2x + 3y = 11$ and $2x - 4y = -24$.
- Q24.** Find the roots of the quadratic equation $2x^2 - 7x + 3 = 0$ by the factorisation method.
- Q25.** In the figure, $DE \parallel BC$. If $AD = 4$ cm, $DB = 6$ cm, $AE = 5$ cm, find EC . (Use BPT.)

SECTION C — Short Answer-II ($6 \times 3 = 18$ marks)

Answer with full working.

- Q26.** Prove that $\sqrt{5}$ is irrational.
- Q27.** Divide the polynomial $p(x) = x^3 - 3x^2 + 5x - 3$ by $g(x) = x^2 - 2$ and find the quotient and the remainder.
- Q28.** A boat goes 30 km upstream and 44 km downstream in 10 hours. In 13 hours it can go 40 km upstream and 55 km downstream. Find the speed of the stream and that of the boat in still water.
- Q29.** Find two consecutive positive odd integers whose product is 323.
- Q30.** Find the sum of the first 22 terms of the AP: 8, 3, -2 , ...
- Q31.** In $\triangle ABC$, if AD is the median, prove that $AB^2 + AC^2 = 2AD^2 + 2(BD)^2$. (Apollonius theorem)

SECTION D — Long Answer ($4 \times 5 = 20$ marks)

Answer in detail with neat figures where required.

- Q32.** Use Euclid's division algorithm to find the HCF of 4052 and 12576. Hence express the HCF as $4052m + 12576n$ for some integers m and n .
- Q33.** Solve the following pair of equations graphically: $x + 2y = 6$ and $2x - y = 2$. Also find the area of the triangle formed by these lines and the x -axis.
- Q34.** The sum of the reciprocals of Rehman's ages (in years) 3 years ago and 5 years from now is $\frac{1}{3}$. Find his present age.
- Q35.** State and prove the Basic Proportionality Theorem (BPT). Hence, in $\triangle ABC$, if D and E are points on AB and AC respectively such that $\frac{AD}{DB} = \frac{AE}{EC} = \frac{2}{3}$ and $DE = 4$ cm, find BC .

SECTION E — Case-based / Source-based ($3 \times 4 = 12$ marks)

Read each case carefully and answer the sub-parts. Each case study has 4 marks.

Q36. Case Study (Real Numbers). A bakery sells two types of biscuit packets: Type A in packs of 24 and Type B in packs of 36. The owner wants to pack them into larger boxes such that each box contains the same number of biscuits and only one type per box, with no biscuit left over.

- What is the maximum number of biscuits that can be put into each box? (1)
- How many boxes of Type A will be required if total Type A biscuits are 240? (1)
- Express the HCF found in (i) as a linear combination of 24 and 36. (2)

OR

- If $\text{LCM}(24, 36)$ is to be calculated, find it and verify $\text{LCM} \times \text{HCF} = 24 \times 36$. (2)

Q37. Case Study (Quadratic Equations / Polynomials). A rectangular plot has length 5 m more than its breadth. Its area is 750 m^2 .

- If breadth = x metres, write the equation in x . (1)
- Find the value of x . (2)
- Find the perimeter of the plot. (1)

Q38. Case Study (Triangles). Suresh wants to find the height of a tree. He stands at point A and observes that the shadow of the tree just reaches up to point A. The shadow of his 1.5 m tall friend, standing at the same time at the same place near the tree, is 1 m long. The shadow of the tree is 8 m long.

- Are the two triangles formed similar? Why? (1)
- Set up the proportion to find the height of the tree. (1)

(iii) Find the height of the tree. (2)

— End of Question Paper —